TSA 2017

Problem Solving

Questions taken from the TSA Oxford 2017 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2017 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) A café owner makes 50 mini pizzas and 50 flapjacks for sale during the day. The pizzas are sold for $2.00 each and the flapjacks are sold for $1.00 each. However, by the end of lunchtime there are 10 mini pizzas still unsold and 15 flapjacks unsold, so the price of each is halved for the afternoon. At the end of the day there are still 2 mini pizzas and 1 flapjack left which are given away free to the last customer.

If the original 50 mini pizzas and 50 flapjacks had a combined production cost of $30.00, how much profit is made on the sale of the mini pizzas and flapjacks for the day?

A $100.00

B $102.50

C $115.00

D $120.00

E $130.00

2 (Q6) A picture 40 cm high by 30 cm wide is to be framed. There will be a mount between the edge of the picture and the frame. This mount will be 6 cm wide at the top and sides, and 9 cm wide at the bottom. The width of the wood used for the frame is 2 cm.

What is the overall height of the framed picture?

A 46 cm

B 49 cm

C 56 cm

D 59 cm

E 62 cm

3 (Q7) Below is a summary of all the second-hand cars sold by Bargain Cars during 2012 and 2013:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2012	85	83	98	102	118	135	88	97	116	83	125	128
2013	123	106	87	114	90	106	124	113	131	77	92	75

Which month of 2013 saw the biggest increase in sales compared with the previous month?

A April

B June

C July

D September

E November

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4 (Q8) Rosie needs to mow her lawn. She has just bought a new lawn mower, and believes she can mow at a rate of $2\,m^2$ a minute. With her old mower, she could only mow the lawn at a rate of $1\,m^2$ a minute. Her lawn is 10 m by 25 m. There is a patio in one corner which is 2 m by 3 m and which she does not need to mow.

How much time does she save with her new lawn mower?

A 122 minutes

B 125 minutes

C 128 minutes

D 244 minutes

E 250 minutes

5 (Q12) Tasty Crisps are currently running a promotion in which there is one money-off coupon inside every packet. Some coupons are worth 19p, some are worth 12p and others are worth 7p.

My friend tells me that he has at least one of each value and his seven coupons have a total value of exactly £1.

How many of my friend's coupons are worth 19p?

A 1

B 2

C 3

D 4

E 5

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6 (Q13) The following table summarises the results of all the matches played last season in a football league. (The home team's score is always given first.) Each team plays every other team in the league twice; once at home and once away.

		away team									
home team		Amazon	Danube	Ganges	Hudson	Nile	Orinoco	Rhine	Tigris	Volga	Yangtze
	Amazon		2–2	2–1	4–0	1–1	2–1	1–3	2–0	1–2	1–1
	Danube	1–3		1–4	2–1	2–3	0–1	2–1	0–0	2–5	0–1
	Ganges	2–2	3–1		2–0	1–2	2–3	1–3	1–2	3–1	2–0
	Hudson	1–2	0–1	2–1		0–1	1–3	2–1	1–0	1–4	0–0
	Nile	2–0	3–0	1–2	3–1		1–0	3–3	2–3	2–0	1–2
	Orinoco	4–1	0–0	3–1	2–0	3–2		2–2	3–1	1–2	2–0
	Rhine	2–3	2–1	1–1	3–0	1–2	2–0		3–0	2–2	3–1
	Tigris	1–1	2–2	1–3	1–0	2–0	0–2	0–3		2–4	1–0
	Volga	3–2	3–0	2–1	4–1	2–2	1–0	1–2	4–0		1–1
	Yangtze	0–2	1–2	2–0	2–1	3–1	0–3	0–2	2–0	1–3	

Which was the only team that scored at least one goal in every match they played last season?

A Amazon

B Ganges

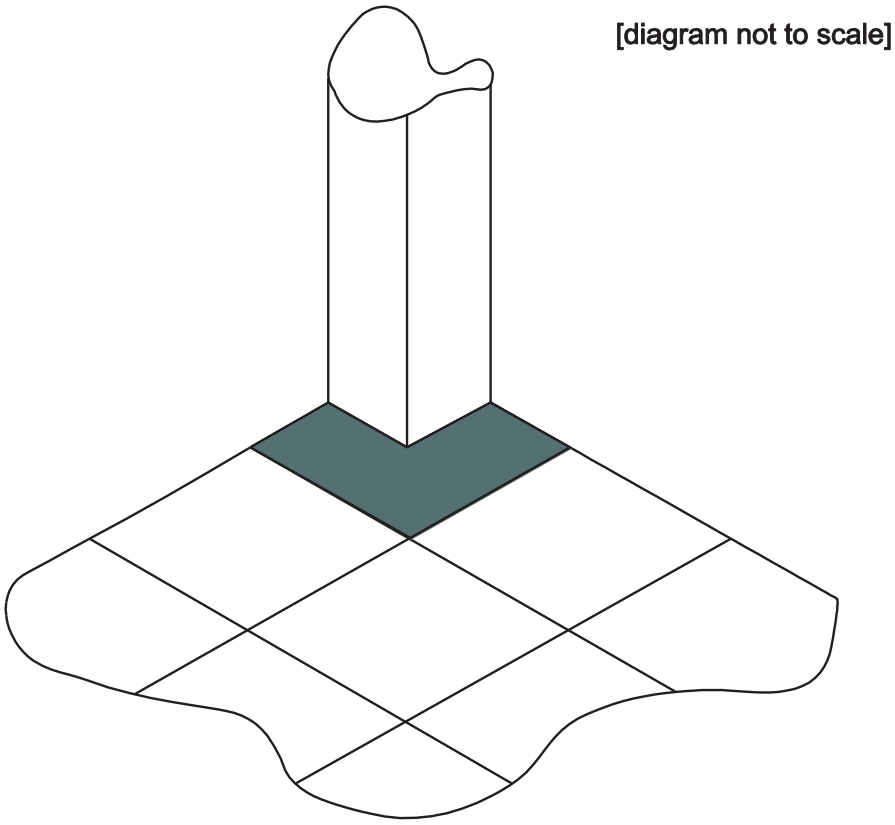
C Nile

D Rhine

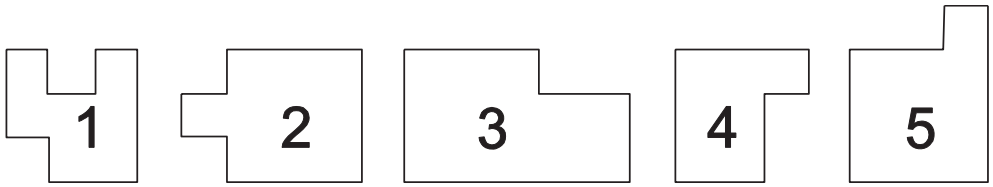
E Volga

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7 (Q14) A tiler has encountered a problem when tiling a kitchen floor. He is one tile short of completing the job. This is indicated by the greyed-out area in the diagram below which shows part of the kitchen floor:



There are some odd pieces of tile left over that could be used to complete the job. Three of these, if combined, would fit the area exactly.



Which three pieces of tile could be pieced together to fit into the remaining corner area? (The tiles are shown face up and cannot be laid face down.)

A 1, 2 and 5

B 1, 3 and 5

C 1, 4 and 5

D 2, 3 and 4

E 2, 4 and 5

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8 (Q18) I am planning to repaint the walls and the ceiling of my dining room. The room has the shape of a rectangular box and is 8 m long, 4 m wide and 3 m high. I do not need to paint the door and the three windows, the combined surface area of which is 10 m^2. Paint is sold in 8-litre tins and 1 litre of paint is sufficient for a surface area of 12 m^2.

How many tins of paint do I need to buy?

A 1

B 2

C 6

D 8

E 17

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9 (Q19) The heptathlon is a seven-event athletics competition. In each event the time or distance recorded is converted into a points score. This is the scoring table that is used for the high jump:

performance	*points*	*performance*	*points*	*performance*	*points*	*performance*	*points*
2.19	1498	1.89	1093	1.59	724	1.29	399
2.18	1484	1.88	1080	1.58	712	1.28	389
2.17	1470	1.87	1067	1.57	701	1.27	379
2.16	1456	1.86	1054	1.56	689	1.26	369
2.15	1442	1.85	1041	1.55	678	1.25	359
2.14	1428	1.84	1029	1.54	666	1.24	350
2.13	1414	1.83	1016	1.53	655	1.23	340
2.12	1400	1.82	1003	1.52	644	1.22	331
2.11	1386	1.81	991	1.51	632	1.21	321
2.10	1373	1.80	978	1.50	621	1.20	312
2.09	1359	1.79	966	1.49	610	1.19	302
2.08	1345	1.78	953	1.48	599	1.18	293
2.07	1332	1.77	941	1.47	588	1.17	284
2.06	1318	1.76	928	1.46	577	1.16	275
2.05	1305	1.75	916	1.45	566	1.15	266
2.04	1291	1.74	903	1.44	555	1.14	257
2.03	1278	1.73	891	1.43	544	1.13	248
2.02	1264	1.72	879	1.42	534	1.12	239
2.01	1251	1.71	867	1.41	523	1.11	231
2.00	1237	1.70	855	1.40	512	1.10	222
1.99	1224	1.69	842	1.39	502	1.09	214
1.98	1211	1.68	830	1.38	491	1.08	205
1.97	1198	1.67	818	1.37	481	1.07	197
1.96	1184	1.66	806	1.36	470	1.06	188
1.95	1171	1.65	795	1.35	460	1.05	180
1.94	1158	1.64	783	1.34	449	1.04	172
1.93	1145	1.63	771	1.33	439	1.03	164
1.92	1132	1.62	759	1.32	429	1.02	156
1.91	1119	1.61	747	1.31	419	1.01	149
1.90	1106	1.60	736	1.30	409	1.00	141

Daphne is competing in a heptathlon competition and has just beaten her previous best high jump performance by 18 cm. This has increased her best points score in the high jump by 214.

What is Daphne's new personal-best high jump performance?

A 1.09 metres

B 1.44 metres

C 1.74 metres

D 1.89 metres

E 2.09 metres

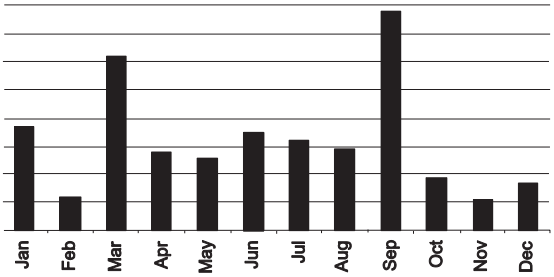
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10 (Q20) After the financial crash there were fewer new car registrations in the UK in 2012 than there were in 2007. The monthly figures for new car registrations for both years are given in the table below.

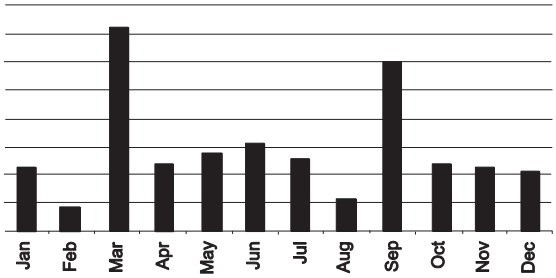
Jan 2012	124.1	Jan 2007	161.2
Feb 2012	61.4	Feb 2007	71.5
Mar 2012	367.9	Mar 2007	445.3
Apr 2012	138.0	Apr 2007	167.9
May 2012	159.0	May 2007	184.8
Jun 2012	185.5	Jun 2007	220.6
Jul 2012	141.7	Jul 2007	175.3
Aug 2012	57.5	Aug 2007	76.2
Sep 2012	359.0	Sep 2007	419.1
Oct 2012	149.7	Oct 2007	168.0
Nov 2012	150.3	Nov 2007	160.5
Dec 2012	123.1	Dec 2007	139.8

Which one of the following charts best represents the monthly differences between new car registrations for these two years?

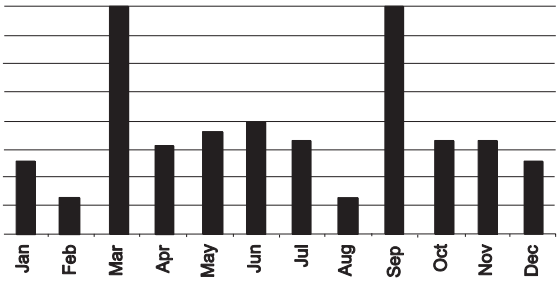
A



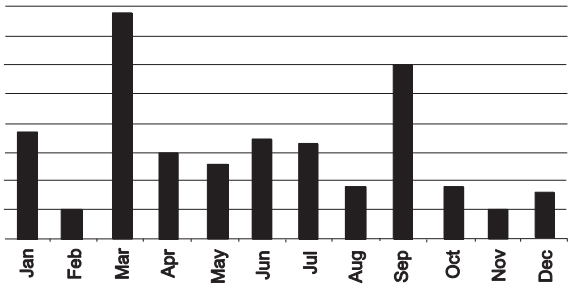
B



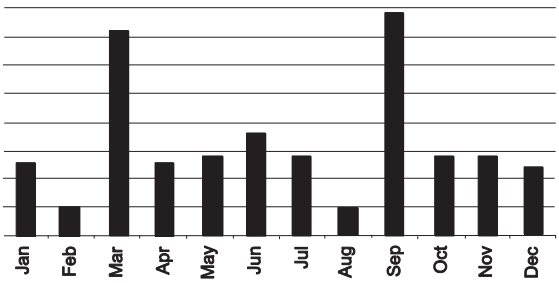
C



D



E



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11 (Q24) A laboratory technician has made up a stock mixture of two chemicals, X and Y, ready for use by all of the groups in Year 9 at a High School. He has mixed 6.0 kg which is $\frac{1}{4}$ of X and $\frac{3}{4}$ of Y. He realises that he has used the wrong mix and, rather than waste the chemicals he has used so far, he is going to add some more of chemical X so that the mix consists of 40% of X and 60% of Y.

What mass of chemical X must he add?

A 1.0 kg

B 1.5 kg

C 2.0 kg

D 2.5 kg

E 3.0 kg

12 (Q25) Five friends wanted to go on holiday together, but they found it difficult to decide where to go. Jo and Mel both wanted to go to Portugal, with France as their second choice. Kim wanted to go to Majorca, while Lexy's first choice was Greece and Naz's was Tenerife. Naz and Kim both put France as their second choice, while Lexy gave it as her third choice, preferring Tenerife as her second choice. Kim and Mel put Tenerife as their third choice. Jo's third choice was Greece, and Naz's was Majorca.

They agreed to allocate three points for a 1st choice, two points for a 2nd choice and one point for a 3rd choice. The friends decided that they would not go anywhere which was not someone's first choice.

Where did they go on holiday?

A France

B Greece

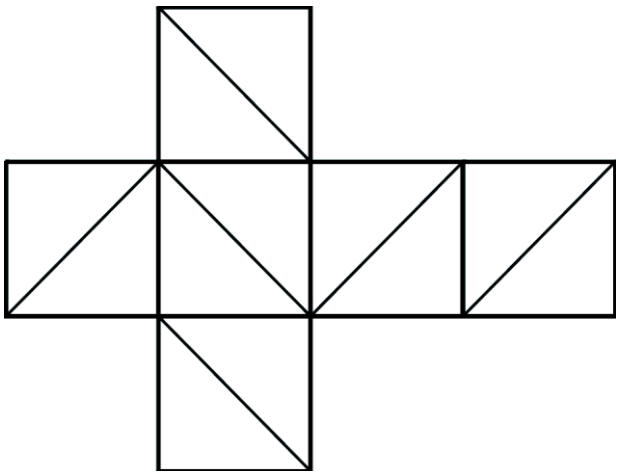
C Majorca

D Portugal

E Tenerife

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13 (Q26) The following net is folded up to make a cube.



At how many corners of the cube do exactly two of the diagonal lines on the faces meet?

A 2

B 3

C 4

D 5

E 6

14 (Q30) There are two ways of scoring points in a ball game: a ‘major’ scores 5 points and a ‘minor’ scores 3 points.

In a match played yesterday, the Reds beat the Blues 77–52 despite the fact that the Blues scored exactly twice as many majors as minors, whilst the Reds scored exactly half as many majors as minors.

How many majors were scored altogether in yesterday’s match?

A 9

B 12

C 15

D 18

E 21

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15 (Q31) The following ingredient and nutrition information (rounded to the nearest 0.1 g or 1 kcal) appears on a 300 g packet of oatcakes.

Each oatcake provides:

energy	53 kcal
fat	2.2 g
saturated fat	0.6 g
salt	0.2 g
sugars	0.3 g

Ingredients: oatmeal (77%), vegetable oils, wheat flour, sugar, salt, raising agent: sodium hydrogen carbonate.

Typical values per 100 g: energy 1792 kJ, 427 kcal; protein 10.7 g; carbohydrate 56.45 g of which sugars 2.4 g, starch 53.7 g; fat 17.6 g of which saturates 5.0 g, mono-unsaturates 7.9 g, polyunsaturates 4.5 g; fibre 8.9 g; salt 1.4 g of which sodium 0.6 g.

How many oatcakes are there in a packet?

A 7

B 8

C 21

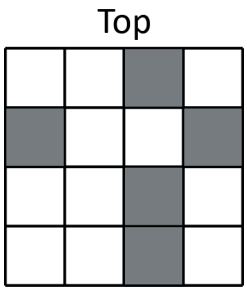
D 24

E 27

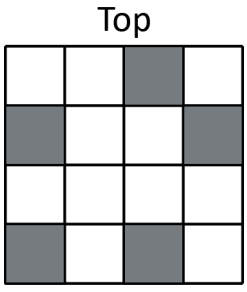
16 (Q32) Instructions were given to shade 5 particular squares on a grid of 16 squares. One student shaded in the squares as instructed, but the other four got exactly 1 square wrong.

Which one of the five grids below shows the correct shading?

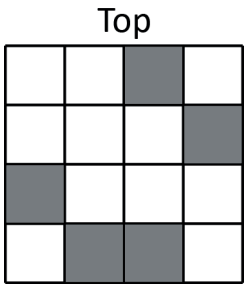
A



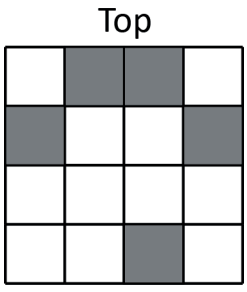
B



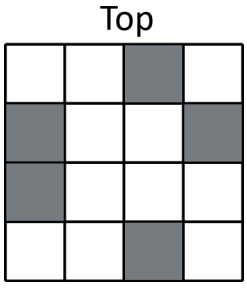
C



D



E



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17 (Q36) Fergus Castle is a tourist attraction on a small island off the coast of the mainland. A ferry service runs between the mainland and the island, and takes 40 minutes to make a single journey in either direction. There is a turnaround time of 5 minutes at the end of each single journey. Each day, ferries leave the mainland every 20 minutes, at 5, 25 and 45 minutes past each hour, starting at 09:05. Ferries leave the island at 15, 35 and 55 minutes past each hour, starting at 09:15.

What is the minimum number of boats needed?

A 3

B 4

C 5

D 6

E 7

18 (Q37) The table below shows the numbers of pupils and teachers in all UK schools in 2003:

	pupils (thousands)	*teachers (thousands)*
England North	2,418	132
England Midlands	1,613	88
England East	900	50
England London	1,173	66
England South	2,160	121
Wales	497	26
Scotland	893	59
Northern Ireland	341	21

Which region had the lowest ratio of pupils per teacher?

A England Midlands

B England South

C Wales

D Scotland

E Northern Ireland

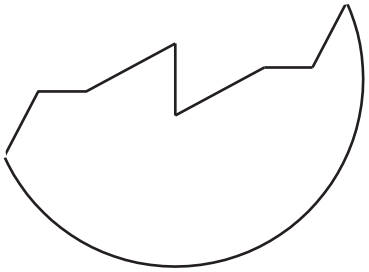
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19 (Q38) When a mixed doubles tournament is held at my tennis club, partners are chosen by the following method.

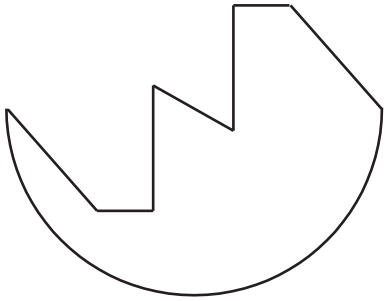
The appropriate number of tokens, all with a different shape, are placed into a box, and the male participants all pick one out. Each token has an identical 'twin' in a second box, from which the female participants all select one. Each pair of identical tokens fit together to make a complete disc. Couples with identical tokens partner each other in the tournament.

Which one of the following can **not** be a token used by my tennis club?

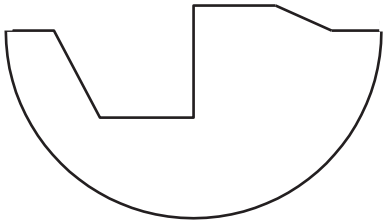
A



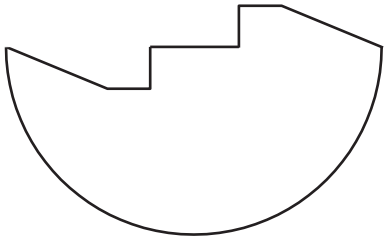
B



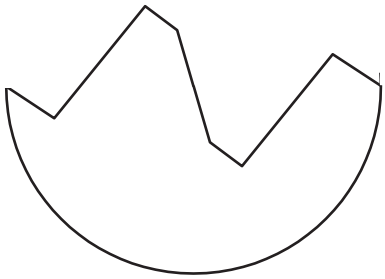
C



D



E



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20 (Q42) Joan has a cat called Tibber.

Every day Tibber is fed 2 sachets of wet food and 25 grams of dry food.

For some time, Joan has bought the cat food from the local pet shop where one box of 12 sachets costs £12.00 and one 400 gram packet of dry food costs £4.00.

Joan has now decided to buy all of the cat food from an online distributor.

Four boxes, each of 24 sachets of wet food, will cost £62.40, and one 2 kilogram packet of dry food will cost £16.00.

How much money will Joan save each day on cat food when she buys it from the online distributor rather than the local pet shop?

A 40p

B 75p

C 93p

D £1.50

E £2.25

21 (Q43) A particular computer game involves the capturing of three types of mythical creatures: Arps, Orps and Urps.

Arps have 6 legs, 3 horns and a tail.
Orps have 4 legs and 2 horns, but no tail.
Urps have 3 legs and a tail, but no horns.

The last time Billy played this game he captured 45 creatures with a total of 222 legs, 99 horns and 33 tails.

How many of the 45 creatures that Billy captured were Urps?

A 8

B 12

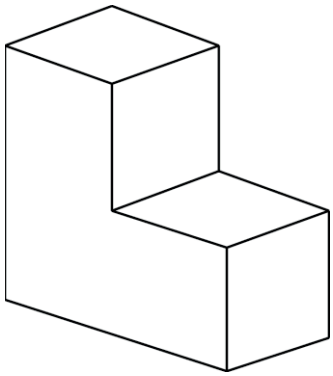
C 15

D 20

E 25

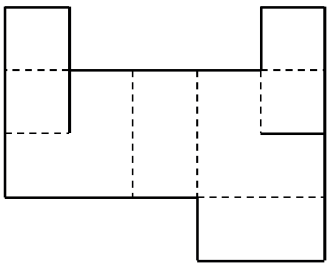
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22 (Q44) The drawing below shows an 'L' shaped box.

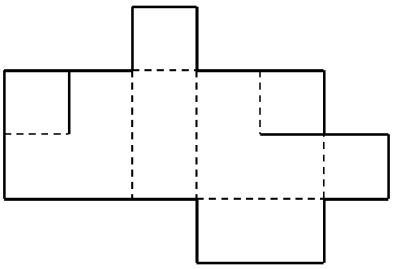


Which one of the nets below **cannot** be folded up to form the 'L' shaped box? (A dotted line indicates a fold and a solid line indicates a cut.)

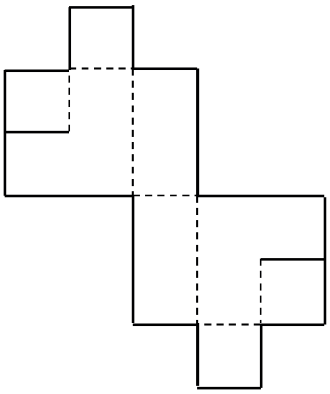
A



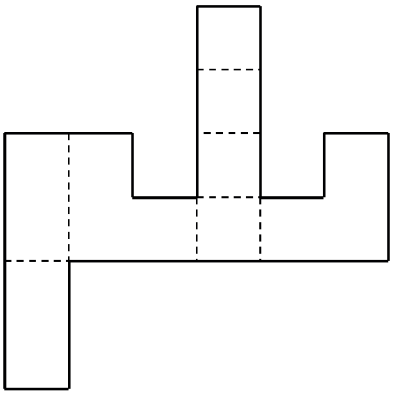
B



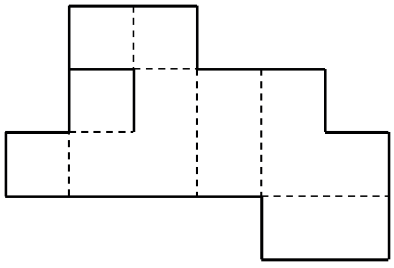
C



D



E



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23 (Q48) Each word in a word game is scored by adding up the values of its letters. Each letter has the same value whenever it appears but different letters have different values. I know the word values for TEAR, RITE, TREE and RAT, but none of the letter values.

How many of the letter values can I now calculate?

A 0

B 1

C 2

D 3

E 4

24 (Q49) Nathan lives in London. His older brother Mark lives in San Diego, his younger brother Ben lives in Barcelona and his sister Isabel lives in Nairobi. Nathan is trying to arrange an online call between all four siblings. Each of the siblings has agreed to be available between 08:00 and 20:00 local time.

San Diego is 8 hours behind London time.
Barcelona is 1 hour ahead of London time.
Nairobi is 2 hours ahead of London time.

For how many hours during a 24-hour period are at least three of the four siblings able to be online at the same time?

A 8

B 9

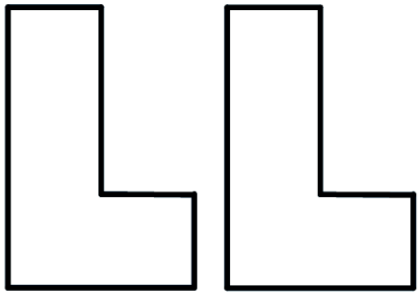
C 10

D 11

E 12

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25 (Q50) Marina purchased a pair of L-shaped tables thinking that they would add an unusual touch to her conference room.

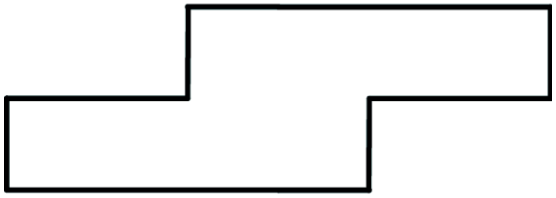


Which one of the following arrangements could **not** be made using the two L-shaped tables?

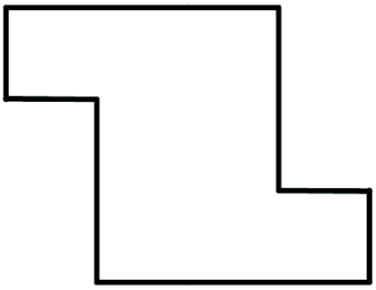
A



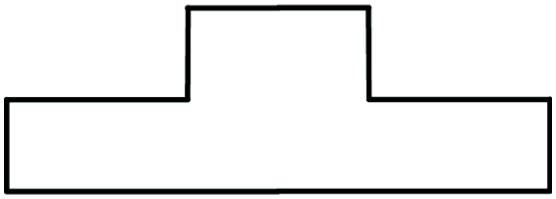
B



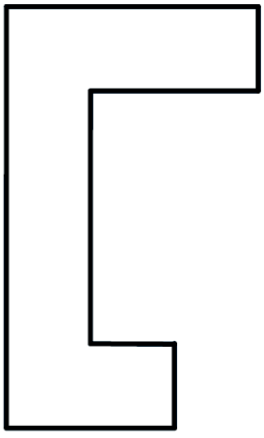
C



D



E



END OF TEST

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